

1 Assignment 1

1. What is the time for light to cross a proton?

Solution:

considering the diameter of proton $d = 1.6 \times 10^{-15}$ m

$$\begin{aligned} t &= \frac{d}{c} \\ &= \frac{1.6 \times 10^{-15}}{3 \times 10^8} \\ &= 5.33 \times 10^{-24} \text{s} \end{aligned}$$

2. Using the property of the metric tensor, show that:

$$A \cdot B = A^0 B^0 - \vec{A} \cdot \vec{B}$$

Solution:

$$\begin{aligned} A \cdot B &= g_{\mu\nu} A^\mu B^\nu \\ &= A^0 B^0 - \delta_{ij} A^i B^j \\ &= A^0 B^0 - \vec{A} \cdot \vec{B} \end{aligned}$$

3. Show that:

$$\Lambda^\alpha{}_\mu \Lambda^\beta{}_\nu \eta_{\alpha\beta} = \eta_{\mu\nu}$$

is a necessary condition for a Lorentz transformation.

Solution:

In Lorentz transformation $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$ is invariant hence

$$\begin{aligned} \eta_{\mu\nu} dx^\mu dx^\nu &= \eta_{\mu\nu} dx'^\mu dx'^\nu \\ &= \eta_{\mu\nu} \Lambda^\mu{}_\sigma \Lambda^\nu{}_\rho dx^\sigma dx^\rho \end{aligned}$$

and therefore

$$\boxed{\eta_{\mu\nu} = \eta_{\rho\sigma} \Lambda^\rho{}_\mu \Lambda^\sigma{}_\nu}$$

4. Relativistic wave equations contain derivatives with respect to space-time. How does a derivative with respect to space-time transform under a Lorentz transformation?

Solution:

The covariant vector transforms according to

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}$$

in Lorentz transformation

derivative with respect to spacetime ∂_{μ} is a covariant vector so they transform as

$$\partial'_{\mu} = \Lambda_{\mu}^{\nu} \partial_{\nu}$$

where $\Lambda_{\mu}^{\nu} = (\Lambda^{\mu}_{\nu})^{-1}$

5. A particle of mass M , initially at rest, decays into two pieces, each of mass m . What is the speed of each particle?

Solution:

Using momentum conservation we'll have momenta of both particles to be same

Now energy conservation will give

$$\begin{aligned} Mc^2 &= 2\gamma mc^2 \\ \frac{1}{\sqrt{1 - v^2/c^2}} &= \frac{M}{2m} \\ \frac{v^2}{c^2} &= 1 - \frac{4m^2}{M^2} \end{aligned}$$

$$\boxed{v = c\sqrt{1 - \frac{4m^2}{M^2}}}$$

6. Consider the decay $X \rightarrow a + b$ where b is massless. What is the speed of a ?

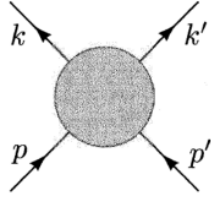
Solution:

Let mass of X be M and that of a be m . Assume particle X initially at rest, therefore both a and b will have same momentum $\mathbf{p}$

Using Energy Conservation

$$\begin{aligned}
M &= \sqrt{p^2 + m^2} + p \\
(M - p)^2 &= p^2 + m^2 \\
M^2 - 2Mp &= m^2 \\
2M\gamma mv &= M^2 - m^2 \\
\frac{v^2}{1 - v^2} &= \left(\frac{M^2 - m^2}{2Mm} \right)^2 \\
v &= \frac{\frac{M^2 - m^2}{2Mm}}{\sqrt{1 + \left(\frac{M^2 - m^2}{2Mm} \right)^2}} \\
v &= \frac{M^2 - m^2}{\sqrt{4M^2m^2 + (M^2 - m^2)^2}} \\
\boxed{v = \frac{M^2 - m^2}{M^2 + m^2}}
\end{aligned}$$

7. Show that the Mandelstam variables s , t , u are invariant.



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$$\begin{aligned}
s &= (p + p')^2 \\
&= (p + p')_\mu (p + p')^\mu \\
&= p^\mu p_\mu + p'^\mu p'_\mu + 2p^\mu p'_\mu \\
&= m_p^2 + m_{p'}^2 + 2p^\mu p'_\mu
\end{aligned}$$

All m_p^2 , $m_{p'}^2$ and $p^\mu p'_\mu$ are Lorentz Scalars, hence are invariant. Thus s is Lorentz invariant

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$$\begin{aligned}
t &= (k - p)^2 \\
&= m_k^2 + m_p^2 - 2k^\mu p_\mu
\end{aligned}$$

Hence Lorentz Invariant

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$$\begin{aligned} u &= (k' - p)^2 \\ &= m_{k'}^2 + m_p^2 - 2k'^\mu p_\mu \end{aligned}$$

Again Lorentz Invariant

8. Show that in a $2 \rightarrow 2$ scattering process:

$$s + t + u = \sum_i m_i^2$$

Solution:

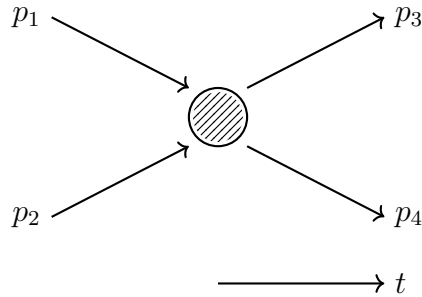


Figure 1:

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2 \\ t &= (p_3 - p_1)^2 = (p_4 - p_2)^2 \\ u &= (p_4 - p_1)^2 = (p_3 - p_2)^2 \end{aligned}$$

$$\begin{aligned} s + t + u &= p_1^2 + p_2^2 + p_3^2 + p_1^2 + p_4^2 + p_1^2 + 2p_1 \cdot p_2 - 2p_1 \cdot p_3 - 2p_1 \cdot p_4 \\ &= 3p_1^2 + p_2^2 + p_3^2 + p_4^2 + 2p_1 \cdot (p_2 - p_3 - p_4) \end{aligned}$$

now by conservation of four momentum

$$\begin{aligned} p_1 + p_2 &= p_3 + p_4 \\ p_2 - p_3 - p_4 &= -p_1 \end{aligned}$$

therefore,

$$\begin{aligned} s + t + u &= 3p_1^2 + p_2^2 + p_3^2 + p_4^2 - 2p_1^2 \\ &= p_1^2 + p_2^2 + p_3^2 + p_4^2 \end{aligned}$$

$$\boxed{s + t + u = \sum_i m_i^2}$$

9. Show that when all masses are neglected in a $2 \rightarrow 2$ reaction, then:

$$t = -\frac{s}{2}(1 - \cos \theta), \quad u = -\frac{s}{2}(1 + \cos \theta)$$

where θ is the scattering angle.

Solution:

Scattering Angle is angle between incoming particle and outgoing particle in center of mass frame

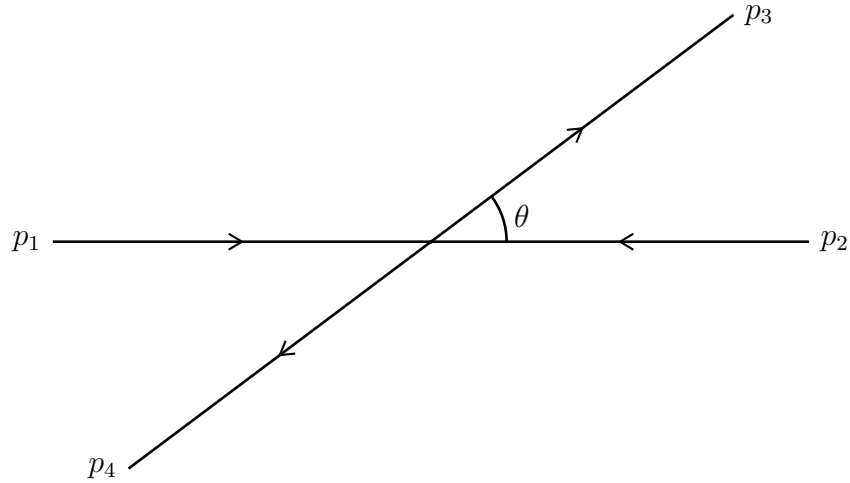


Figure 2: 2-2 Scattering in Center of Mass Frame

$$p_1 = (E, E, 0, 0, 0)$$

$$p_2 = (E, -E, 0, 0)$$

$$p_3 = (E, E \cos \theta, E \sin \theta, 0)$$

$$p_4 = (E, -E \cos \theta, -E \sin \theta, 0)$$

Now,

$$\begin{aligned}
 t &= (p_1 - p_3)^2 \\
 &= -2p_1 \cdot p_3 \quad (\text{since } m_i = 0) \\
 &= -2(E^2 - E^2 \cos \theta) \\
 &= -2E^2(1 - \cos \theta)
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 s &= (p_1 + p_2)^2 \\
 &= 4E^2 \\
 E^2 &= \frac{s}{4}
 \end{aligned}$$

Hence,

$$t = -\frac{s}{4}(1 - \cos \theta)$$

In similar way we can show

$$\begin{aligned}
 u &= (p_1 - p_4)^2 \\
 &= -2p_1 \cdot p_4 \\
 &= -2E^2(1 + \cos \theta)
 \end{aligned}$$

$$u = -\frac{s}{4}(1 + \cos \theta)$$

10. A particle of mass M , initially at rest, decays into two particles, each of mass m . What is the speed of each as they fly off?

Solution:

Same question as question 5

11. Show that in the rest frame of a decaying particle $a \rightarrow b + c$:

$$E_b = \frac{M_a^2 - M_c^2 + M_b^2}{2M_a}$$

Solution:

we write four-momentum of particles as

$$p_a = (M_a, \mathbf{0}) \quad p_b = (E_b, \mathbf{p}) \quad p_c = (E_c, -\mathbf{p})$$

also,

$$E_b^2 = \mathbf{p}^2 + M_b^2 \tag{1}$$

$$M_a = E_b + E_c \quad (2)$$

Now

$$\begin{aligned} p_a^2 &= p_b^2 + p_c^2 + 2p_a \cdot p_b \\ M_a^2 &= M_b^2 + M_c^2 + 2(E_b E_c + \mathbf{p}^2) \\ &= M_b^2 + M_c^2 + 2(E_b(M_a - E_b) + E_b^2 - M_b^2) \quad (\text{using (1) and (2)}) \\ &= -M_b^2 + M_c^2 + 2M_a E_b \end{aligned}$$

$$\boxed{E_b = \frac{M_a^2 - M_c^2 + M_b^2}{2M_a}} \quad (3)$$

12. Suppose two identical particles, each of mass m and kinetic energy T , collide head-on. Find the kinetic energy T' in the fixed-target frame of reference.

Solution:

Let particles collide head on. To find T' consider the transformation of E in Lorentz transformation with boost such that one particle is at rest

$$E' = \gamma(E - p^1 v^1)$$

We also have relations

$$\begin{aligned} p &= \sqrt{T^2 + 2Tm} \\ v &= \frac{p}{E} = \frac{\sqrt{T^2 + 2Tm}}{T + m} \end{aligned}$$

Therefore,

$$\begin{aligned} T' + m &= \frac{T + m + \sqrt{T^2 + 2Tm} \times \frac{\sqrt{T^2 + 2Tm}}{T + m}}{\sqrt{1 - \frac{T^2 + 2Tm}{(T + m)^2}}} \\ &= \frac{(T + m)^2 + T^2 + 2Tm}{m} \\ &= \frac{2T^2 + 4Tm}{m} \end{aligned}$$

$$\boxed{T' = \frac{2T(T + 2m)}{m}}$$

13. Consider the decay $\pi^+ \rightarrow \mu^+ \nu_\mu$, where $E_{\pi^+} = 20 \text{ GeV}$ and $\tau_{\pi^+} = 2.6 \times 10^{-8} \text{ s}$. What is the time dilation?

Solution:

We have

$$\gamma = \frac{E}{m_\mu}$$

Mass of Muon $m_\mu \approx 106 \text{ MeV}$

$$\gamma = \frac{20000}{106} = 188.68$$

Therefore the time dilated

$$t' = 2.6 \times 10^{-8} \times 188.68 = 4.9$$

$$\boxed{t' = 4.9 \text{ } \mu\text{s}}$$

Time is dilated by factor of ~ 188

14. In the scattering process $a + b \rightarrow c + d$, where θ^* is the scattering angle in the CMS, show that:

$$\cos \theta^* = \frac{s(t - u) + (m_a^2 - m_b^2)(m_c^2 - m_d^2)}{\lambda^{1/2}(s, m_a^2, m_b^2) \lambda^{1/2}(s, m_c^2, m_d^2)}$$

And for elastic scattering, show that:

$$t = -4|\vec{p}|^2 \sin^2 \left(\frac{\theta^*}{2} \right)$$

where $\vec{p}$ is the initial momentum in the CMS

Solution:

We'll use the relations

$$E_a = \frac{s + m_a^2 - m_b^2}{2\sqrt{s}} \quad E_b = \frac{s + m_b^2 - m_a^2}{2\sqrt{s}}$$

$$|\mathbf{p}| = \frac{\lambda^{1/2}(s, m_a^2, m_b^2)}{2\sqrt{s}} \quad |\mathbf{p}'| = \frac{\lambda^{1/2}(s, m_c^2, m_d^2)}{2\sqrt{s}}$$

$$\begin{aligned} t &= (p_a - p_c)^2 \\ &= m_a^2 + m_c^2 - 2(E_a E_c - |\mathbf{p}| |\mathbf{p}'| \cos \theta) \end{aligned}$$

$$\begin{aligned} u &= (p_a - p_d)^2 \\ &= m_a^2 + m_d^2 - 2(E_a E_d + |\mathbf{p}| |\mathbf{p}'| \cos \theta) \end{aligned}$$

$$t - u = m_c^2 - m_d^2 - 2E_a(E_c - E_d) + 4|\mathbf{p}| |\mathbf{p}'| \cos \theta$$

$$E_c - E_d = \frac{m_c^2 - m_d^2}{\sqrt{s}}$$

$$\begin{aligned} t - u &= m_c^2 - m_d^2 - 2 \left(\frac{s + m_a^2 - m_b^2}{2\sqrt{s}} \right) \left(\frac{m_c^2 - m_d^2}{\sqrt{s}} \right) + 4 \frac{\lambda^{1/2}(s, m_a^2, m_b^2) \lambda^{1/2}(s, m_c^2, m_d^2)}{4s} \cos \theta \\ s(t - u) &= s(m_c^2 - m_d^2) - (s + m_a^2 - m_b^2) (m_c^2 - m_d^2) + \lambda^{1/2}(s, m_a^2, m_b^2) \lambda^{1/2}(s, m_c^2, m_d^2) \cos \theta \\ s(t - u) &= - (s - s + m_a^2 - m_b^2) (m_c^2 - m_d^2) + \lambda^{1/2}(s, m_a^2, m_b^2) \lambda^{1/2}(s, m_c^2, m_d^2) \cos \theta \\ s(t - u) &= - (m_a^2 - m_b^2) (m_c^2 - m_d^2) + \lambda^{1/2}(s, m_a^2, m_b^2) \lambda^{1/2}(s, m_c^2, m_d^2) \cos \theta \end{aligned}$$

$$\boxed{\cos \theta = \frac{s(t - u) + (m_a^2 - m_b^2) (m_c^2 - m_d^2)}{\lambda^{1/2}(s, m_a^2, m_b^2) \lambda^{1/2}(s, m_c^2, m_d^2)}}$$

For Elastic Scattering $a = c$ and $b = d$, therefore, $|\mathbf{p}| = |\mathbf{p}'|$

$$\begin{aligned} t &= (p_a - p_c)^2 \\ &= (E_a - E_c)^2 - (\mathbf{p}_a - \mathbf{p}_c)^2 \\ &= -2(|\mathbf{p}|^2 - |\mathbf{p}|^2 \cos \theta) \end{aligned}$$

$$\boxed{t = -4|\mathbf{p}|^2 \sin^2 \left(\frac{\theta}{2} \right)}$$

15. The decay width of the η -meson is $\Gamma_\eta = 2.6$ keV. What is the lifetime τ in seconds?

Solution:

$$\begin{aligned} \tau &= \frac{1}{\Gamma} \\ &= \frac{1}{2.6 \text{ keV}} \\ &= 0.3846 (\text{keV})^{-1} \end{aligned}$$

$$\begin{aligned} 1 \text{ s} &= 1.52 \times 10^{24} (\text{GeV})^{-1} \\ &= 1.52 \times 10^{18} (\text{keV})^{-1} \\ (\text{keV})^{-1} &= 6.58 \times 10^{-19} \text{ s} \\ \tau &= 0.3846 \times 6.58 \times 10^{-19} \end{aligned}$$

$$\boxed{\tau = 2.53 \times 10^{-19} \text{ s}}$$

16. The lifetime of π^0 is $\tau_{\pi^0} = 0.89 \times 10^{-16}$ s. What is the decay width Γ in natural units?

Solution:

$$0.89 \times 10^{-16} \text{ s} = 0.1353 \text{ (eV)}^{-1}$$

Decay Width

$$\begin{aligned}\Gamma &= \frac{1}{\tau} \\ &= \frac{1}{0.1353}\end{aligned}$$

$$\boxed{\Gamma = 7.4 \text{ eV}}$$

17. Show that $d^4p = dp^0 dp^1 dp^2 dp^3$ is a Lorentz-invariant quantity.

Solution:

d^4p will transform like a tensor-density in Lorentz transformation

$$d^4p' = \left| \frac{\partial x'}{\partial x} \right| d^4p$$

where $\left| \frac{\partial x'}{\partial x} \right|$ is the Jacobian of transformation, which is

$$\det(\Lambda^\mu{}_\nu)$$

Since in a proper orthochronous Lorentz transformation $\det(\Lambda)=1$, the quantity d^4p is Lorentz invariant

18. Consider the hyperon decay $\Lambda \rightarrow p + \pi^-$. Given:

$$M_\Lambda = 1.1157 \text{ GeV}, \quad M_p = 0.9383 \text{ GeV}, \quad m_{\pi^-} = 0.1396 \text{ GeV},$$

calculate the energy of the outgoing pion.

Solution:

We can use eqn. (3) from problem 11

$$E_\pi = \frac{M_\Lambda^2 - M_p^2 + M_\pi^2}{2M_\Lambda}$$

Therefore,

$$E_\pi = \frac{1.1157^2 - 0.9383^2 + 0.1396^2}{2 \times 1.1157}$$

$$\boxed{E_\pi = 0.142 \text{ GeV}}$$

19. Two particles of mass m each and speed v collide at right angles forming a new body of mass M . What is the new mass M ?

Solution:

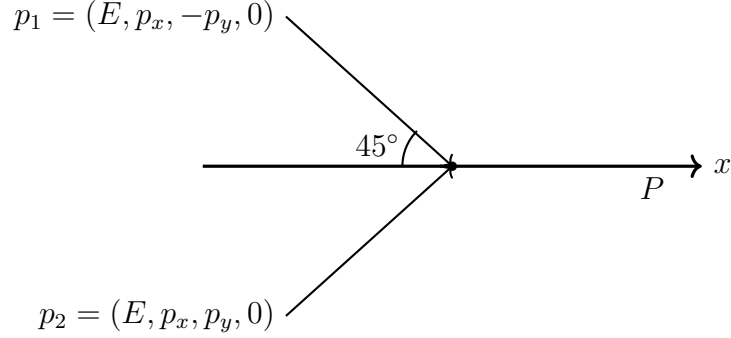


Figure 3:

Use $p^\mu p_\mu = m^2$ we have

$$\begin{aligned}
 p_1^2 + p_2^2 + 2p_1 \cdot p_2 &= M^2 \\
 2m^2 + 2(E^2 - p_x^2 + p_y^2) &= M^2 \\
 2m^2 + 2E^2 &= M^2 \quad \left(\text{since } p_x = p_y = \frac{p}{\sqrt{2}} \right) \\
 \frac{2m^2}{1 - v^2} + 2m^2 &= M^2
 \end{aligned}$$

$$\boxed{M = m \sqrt{2 \left(1 - \frac{1}{1 - v^2} \right)}}$$

20. Show that rapidity η is additive.

Solution:

Relativistically the velocity in same direction is added according to

$$v = \frac{v + v'}{1 + \frac{vv'}{c^2}} \tag{4}$$

Now we define rapidity ξ as

$$\beta = \frac{v}{c} = \tanh \xi$$

we can now write

$$v = c \tanh \xi$$

$$v' = c \tanh \xi'$$

Equation (4) can be written as

$$\begin{aligned} &= \frac{c \tanh \xi + c \tanh \xi'}{1 + \tanh \xi \tanh \xi'} \\ &= c \tanh (\xi + \xi') \end{aligned}$$

which shows that

$$\boxed{\xi_{\text{res}} = \xi + \xi'}$$

and hence rapidity is additive

Particle Physics

Assignment – 2

1. Show that for any two vectors $\vec{A}$ and $\vec{B}$,

$$(\boldsymbol{\sigma} \cdot \vec{A})(\boldsymbol{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B} \mathbf{1} + i\boldsymbol{\sigma} \cdot (\vec{A} \times \vec{B}),$$

where $\boldsymbol{\sigma}$ are the Pauli matrices.

Solution

We'll use the relations

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij}$$

now

$$\begin{aligned} \sigma_i A_i \sigma_j B_j &= \left(\frac{1}{2} \{\sigma_i, \sigma_j\} + \frac{1}{2} [\sigma_i, \sigma_j] \right) A_i B_j \\ &= (\delta_{ij} + i\epsilon_{ijk}\sigma_k) A_i B_j \\ &= A_i B_i + i\sigma_k (\vec{A} \times \vec{B})_k \end{aligned}$$

and hence

$$\boxed{(\boldsymbol{\sigma} \cdot \vec{A})(\boldsymbol{\sigma} \cdot \vec{B}) = \vec{A} \cdot \vec{B} \mathbf{1} + i\boldsymbol{\sigma} \cdot (\vec{A} \times \vec{B})}$$

2. Show that the ratio of the decay widths of

$$\frac{\Gamma(\Delta^{++} \rightarrow p\pi^+)}{\Gamma(\Delta^+ \rightarrow n\pi^+)} = 3,$$

using the Clebsch–Gordon coefficients.

Solution:

We will consider $|I \ I_3\rangle$ basis for particles.

We will consider the reaction rate is

$$\sigma \sim |\text{amplitude}|^2 \sim \sum_I |\langle I' \ I'_3 | A | I \ I_3 \rangle|^2$$

The matrix elements can be calculated using Wigner-Eckart Theorem and will be proportional to Clebsch-Gordon coefficients.

We have $|\Delta^{++}\rangle = \left| \frac{3}{2} \frac{3}{2} \right\rangle$ and $|\Delta^+\rangle = \left| \frac{3}{2} \frac{1}{2} \right\rangle$ since p has $I_3 = 1/2$ and $\pi^+ \ 1$ we get $|p\pi^+\rangle = \left| \frac{3}{2} \frac{3}{2} \right\rangle$.

now since $|\pi^+ n\rangle = |1 - \frac{1}{2}\rangle$ in $|I_1 I_2\rangle$ basis we need to find it's representation in $|I I_3\rangle$

$1 \times 1/2$		$3/2$								
		$+3/2$								
$+1$	$+1/2$	1		$3/2$	$1/2$					
				$+1/2$	$+1/2$					
$+1$	$-1/2$	$1/3$	$2/3$	$3/2$	$1/2$					
0	$+1/2$	$2/3$	$-1/3$	$-1/2$	$-1/2$					
		0	$-1/2$	$2/3$	$1/3$	$3/2$				
		-1	$+1/2$	$1/3$	$-2/3$	$-3/2$				
				-1	$-1/2$	1				

$$\left| \frac{3}{2} \frac{1}{2} \right\rangle = \frac{1}{\sqrt{3}} \left| 1 - \frac{1}{2} \right\rangle + \sqrt{\frac{2}{3}} \left| 0 \frac{1}{2} \right\rangle$$

$$\left| \frac{1}{2} \frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| 1 - \frac{1}{2} \right\rangle - \frac{1}{\sqrt{3}} \left| 0 \frac{1}{2} \right\rangle$$

and therefore

$$\left| 1 - \frac{1}{2} \right\rangle = \sqrt{\frac{2}{3}} \left| \frac{1}{2} \frac{1}{2} \right\rangle + \frac{1}{\sqrt{3}} \left| \frac{3}{2} \frac{1}{2} \right\rangle = |\pi^+ n\rangle$$

now

$$\begin{aligned} \frac{\Gamma(\Delta^{++} \rightarrow p\pi^+)}{\Gamma(\Delta^+ \rightarrow n\pi^+)} &= \left| \frac{\langle \frac{3}{2} \frac{3}{2} | \frac{3}{2} \frac{3}{2} \rangle}{\langle \frac{3}{2} \frac{1}{2} | \left(\sqrt{\frac{2}{3}} \left| \frac{1}{2} \frac{1}{2} \right\rangle + \frac{1}{\sqrt{3}} \left| \frac{3}{2} \frac{1}{2} \right\rangle \right)} \right|^2 \\ &= \left(\frac{1}{\frac{1}{\sqrt{3}}} \right)^2 \end{aligned}$$

and therefore

$$\boxed{\frac{\Gamma(\Delta^{++} \rightarrow p\pi^+)}{\Gamma(\Delta^+ \rightarrow n\pi^+)} = 3}$$

3. Suppose the pion–nucleon interaction Lagrangian is

$$\mathcal{L}_{\text{int}} = g \bar{N} \gamma_5 \tau^a N \pi^a,$$

where g is the coupling constant, τ^a are the Pauli matrices, N is the nucleon field, and $\pi^a = (\pi^1, \pi^2, \pi^3)$. Identify the charged and neutral pions in terms of π^1, π^2, π^3 .

4. Show that the structure constants f^{abc} of the SU(3) Lie algebra can be written as

$$f^{abc} = \frac{1}{4i} \text{Tr}([\lambda^a, \lambda^b] \lambda^c).$$

Solution:

$$\begin{aligned}
\frac{1}{4i} \text{Tr}([\lambda^a, \lambda^b] \lambda^c) &= \frac{1}{2} f^{abd} \text{Tr}(\lambda^d \lambda^c) \\
&= f^{abd} \delta_{dc} \\
&= f^{abc}
\end{aligned}$$

where we have used the fact that $\text{Tr}(\lambda^a \lambda^c) = 2\delta_{ac}$ and $[\lambda^a, \lambda^b] = 2if^{abc}\lambda^c$

5. Show that the Gell–Mann matrices λ^a satisfy

$$\{\lambda^a, \lambda^b\} = C^{ab} \mathbf{1} + 2d^{abc} \lambda^c,$$

where

$$C^{ab} = \frac{4}{3} \delta^{ab},$$

and

$$d^{abc} = \frac{1}{4} \text{Tr}(\{\lambda^a, \lambda^b\} \lambda^c).$$

Solution:

We'll again use the property $\text{Tr}(\lambda^a \lambda^c) = 2\delta_{ac}$. We can write any arbitrary matrix as

$$\begin{aligned}
M &= a_0 \mathbf{1} + a_\alpha \lambda^\alpha \\
&\text{(Put } M \text{ as } \{\lambda^a, \lambda^b\}) \\
\text{Tr}(\{\lambda^a, \lambda^b\}) &= 3a_0 \\
a_0 &= \frac{1}{3} \text{Tr}(\lambda^a \lambda^b + \lambda^b \lambda^a) \\
a_0 &= \frac{2}{3} \text{Tr}(\lambda^a \lambda^b) \\
a_0 &= \frac{4}{3} \delta_{ab}
\end{aligned}$$

Now multiply first equation with λ^c and then take trace

$$\begin{aligned}
\text{Tr}(\{\lambda^a, \lambda^b\} \lambda^c) &= a_0 \text{Tr}(\lambda^c) + a_\alpha \text{Tr}(\lambda^\alpha \lambda^c) \\
\text{Tr}(\{\lambda^a, \lambda^b\} \lambda^c) &= 2a_\alpha \delta_{\alpha c} \\
a_c &= \frac{1}{2} \text{Tr}(\{\lambda^a, \lambda^b\} \lambda^c)
\end{aligned}$$

And hence

$$\boxed{\{\lambda^a, \lambda^b\} = \frac{4}{3} \delta_{ab} \mathbf{1} + \frac{1}{2} \text{Tr}(\{\lambda^a, \lambda^b\} \lambda^c) \lambda^c}$$

6. Using Young tableaux, find the irreducible representations for:

(a) $6 \otimes 3$ of $SU(3)$,

Solution:

6 corresponds to $\square\square$ therefore,

$$\square\square \otimes \square = \begin{array}{|c|c|c|} \hline 3 & 4 & 5 \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline 3 & 4 \\ \hline 2 \\ \hline \end{array}$$

where the dimension of first diagram will be

$$d = \frac{3 \times 4 \times 5}{3 \times 2} = 10$$

and of second diagram

$$d = \frac{4 \times 3 \times 2}{3} = 8$$

therefore

$$\boxed{6 \otimes 3 = 10 \oplus 8}$$

(b) $4 \otimes 4 \otimes 4$ of $SU(4)$,

Solution:

$$\square \otimes \square = \square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$$

Therefore

$$\begin{aligned} & (\square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}) \otimes \square \\ &= \square\square\square \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \\ &= \begin{array}{|c|c|c|} \hline 4 & 5 & 6 \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline 4 & 5 \\ \hline 3 & \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline 4 & 5 \\ \hline 3 & \\ \hline \end{array} \oplus \begin{array}{|c|} \hline 4 \\ \hline 3 \\ \hline 2 \\ \hline \end{array} \end{aligned}$$

The dimension of first diagram is

$$\frac{4 \times 5 \times 6}{2 \times 3} = 20$$

second and third is

$$\frac{4 \times 5 \times 3}{3} = 20$$

and last one is 4 therefore

$$\boxed{4 \otimes 4 \otimes 4 = 20 \oplus 20 \oplus 20 \oplus \bar{4}}$$

(c) $4 \otimes \bar{4}$ of $SU(4)$.

Solution:

$$\square \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|} \hline 1 \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline 4 & 5 \\ \hline 3 & \\ \hline 2 & \\ \hline \end{array}$$

with

$$d_1 = 1$$

and

$$d_2 = \frac{4 \times 5 \times 3 \times 2}{4 \times 2} = 15$$

therefore

$$\boxed{4 \otimes \bar{4} = 1 \oplus 15}$$

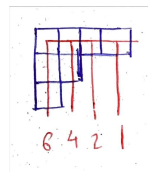
7. Determine the dimension of the irreducible representations shown by the Young tableaux:

(a) $\begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & & \\ \hline \square & & & \\ \hline \end{array}$ of $SU(3)$,

Solution:

$$\begin{array}{|c|c|c|c|} \hline 3 & 4 & 5 & 6 \\ \hline 2 & 3 & & \\ \hline 1 & & & \\ \hline \end{array}$$

multiplying all these numbers we get numerator while finding dimensions
to get the denominator we use hook technique



therefore

$$d = \frac{3 \times 4 \times 5 \times 6 \times 2 \times 3}{6 \times 4 \times 2}$$

$$\boxed{d = 30}$$

(b) $\square \square \square$ of $SU(4)$.

Solution:

$$\begin{array}{|c|c|c|} \hline 4 & 5 & 6 \\ \hline \end{array}$$

We get

$$d = \frac{4 \times 5 \times 6}{2 \times 3}$$

$$d = 20$$

8. Determine the Young tableaux for the direct products:

(a) $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \\ \hline \end{array}$ of SU(3),

Solution:

$$\begin{aligned} & \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|} \hline a \\ \hline b \\ \hline \end{array} = \\ & \left(\begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline & & \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline & \\ \hline & a \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline a \\ \hline \end{array} \right) \otimes \begin{array}{|c|} \hline b \\ \hline \end{array} \\ & = \begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline & & b \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline & & a \\ \hline & & b \\ \hline & & \\ \hline \end{array} \oplus \begin{array}{|c|c|c|c|} \hline & & a & b \\ \hline & & & \\ \hline & & & \\ \hline \end{array} \oplus \\ & \begin{array}{|c|c|} \hline & \\ \hline & a \\ \hline b \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline & & b \\ \hline & & a \\ \hline & & \\ \hline \end{array} \oplus \\ & \begin{array}{|c|c|c|} \hline & & b \\ \hline & & \\ \hline a & & \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline & \\ \hline & b \\ \hline a \\ \hline \end{array} \end{aligned}$$

Now using the rule that: *Reading along the rows from right to left from the top row down to the bottom row, the number of a's must be greater than or equal to the number of b's*, We get final answer

$$\boxed{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \\ \hline \end{array} = \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \\ \hline \\ \hline \end{array}}$$

i.e

$$8 \otimes 3 = 6 \oplus 15 \oplus \bar{3}$$

(b) $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$ of SU(3).

Solution:

$$\begin{aligned}
& \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|c|} \hline a & a \\ \hline \end{array} = \\
& \left(\begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline & \\ \hline & a \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \\ \hline a \\ \hline \end{array} \right) \otimes \begin{array}{|c|} \hline a \\ \hline \end{array} \\
& = \begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline a & & \end{array} \oplus \begin{array}{|c|c|c|c|} \hline & & a & a \\ \hline & & & \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline & & a \\ \hline \end{array} \oplus \\
& \begin{array}{|c|c|} \hline & \\ \hline & a \\ \hline a & \end{array} \oplus \begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline & & a \\ \hline \end{array} \oplus \\
& \begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline a & & \end{array} \oplus \begin{array}{|c|c|} \hline & \\ \hline & a \\ \hline a & \end{array} \\
& = \begin{array}{|c|c|} \hline & a \\ \hline \end{array} \oplus \begin{array}{|c|c|c|c|} \hline & & a & a \\ \hline & & & \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline & & a \\ \hline & & \\ \hline & & a \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \\ \hline a \\ \hline \end{array} \oplus \\
& \begin{array}{|c|c|c|} \hline & a & \\ \hline & & a \\ \hline \end{array}
\end{aligned}$$

Now removing similar diagrams we get final answer

$$\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} \otimes \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \end{array} \oplus \begin{array}{|c|} \hline \\ \hline \\ \hline \end{array} \oplus \begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline & & & \\ \hline & & & \end{array} \oplus \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$$

i.e.

$$8 \otimes 6 = 15 \oplus \bar{3} \oplus 24 \oplus 6$$

9. If $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$ represents the 6 of SU(3), what does $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline a & a \end{array}$ represent?

Solution:

if we add the first diagram just below the second one we get

$$\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline a & a \end{array} = 1$$

i.e. singlet. hence $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$ represent the conjugate representation 6^*

10. Draw the weight diagrams for SU(3):

Name all the particles and their quark compositions if $3 = (u, d, s)$ and determine the corresponding quantum numbers.

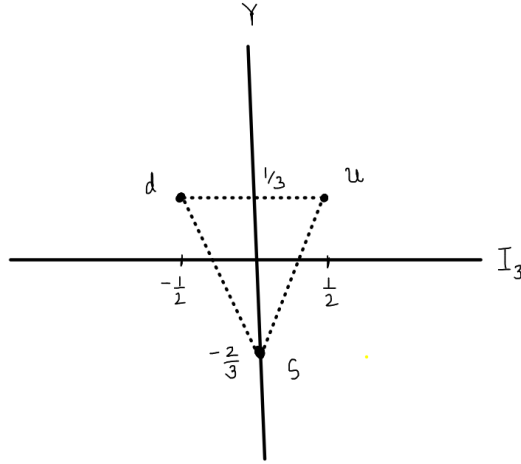
- (a) $3 \otimes 3 \otimes 3$

Solution:

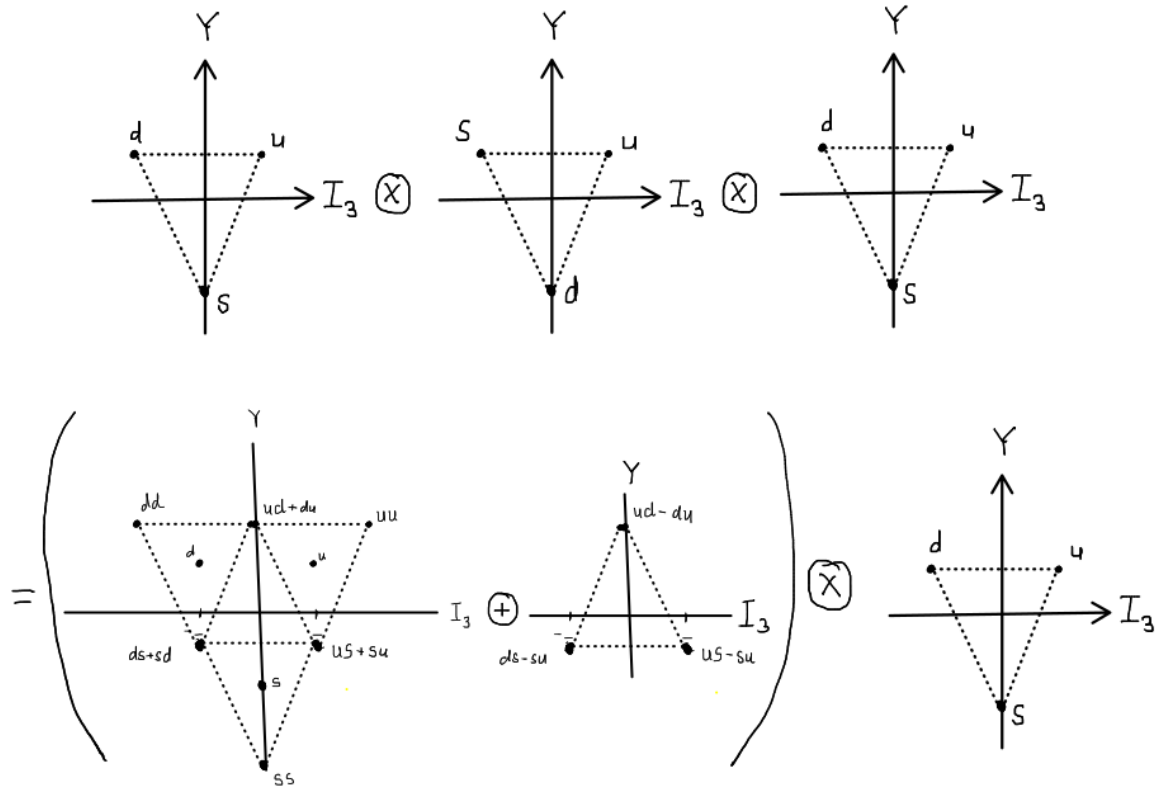
We have two diagonal generators in SU(3):

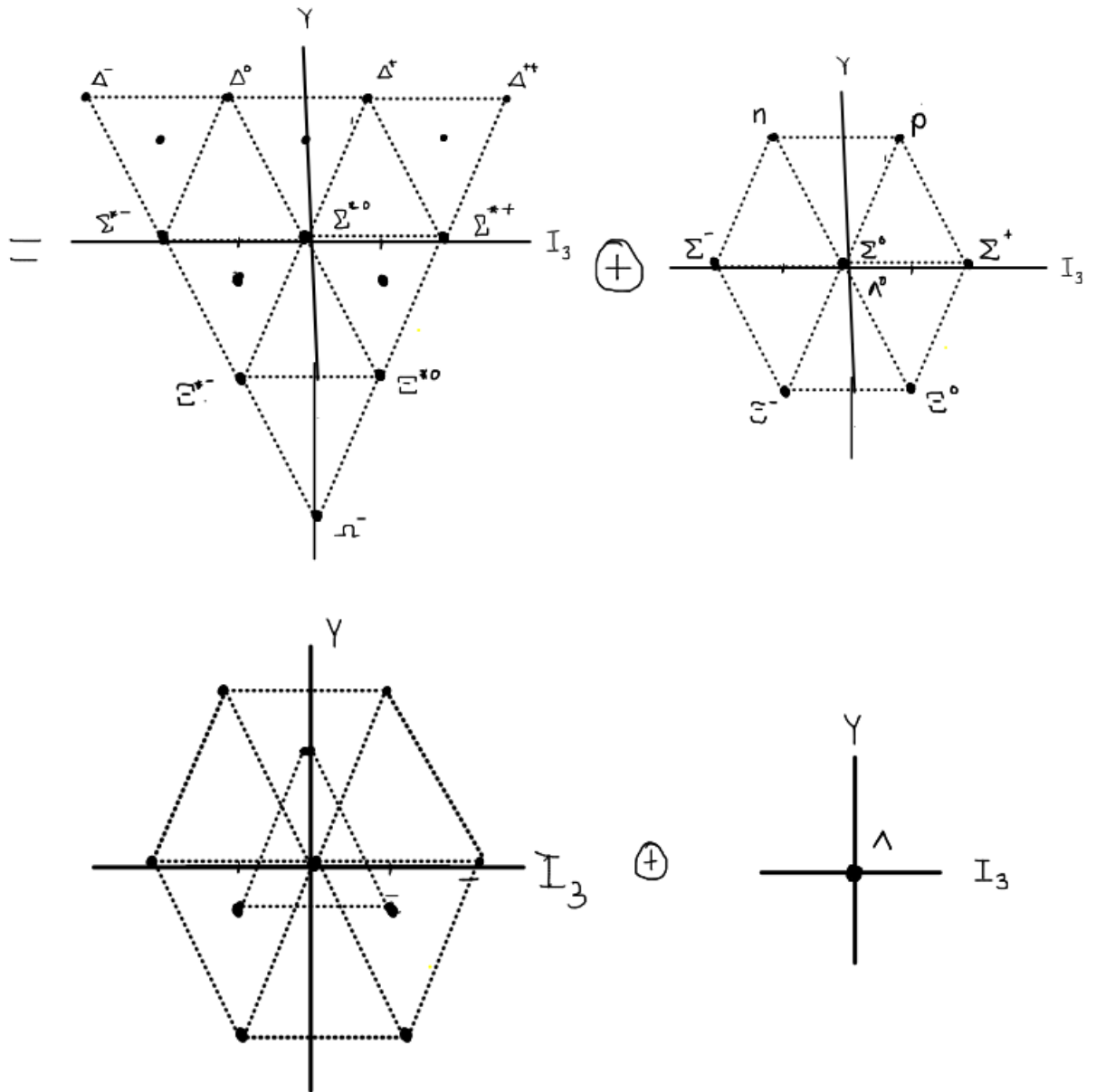
$$\lambda_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda_8 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

Hence we have 3 root vectors $(\frac{1}{2}, \frac{1}{2\sqrt{3}})$, $(-\frac{1}{2}, \frac{1}{2\sqrt{3}})$ and $(0, -\frac{2}{2\sqrt{3}})$. From which we can weight diagram



now for $3 \otimes 3 \otimes 3$

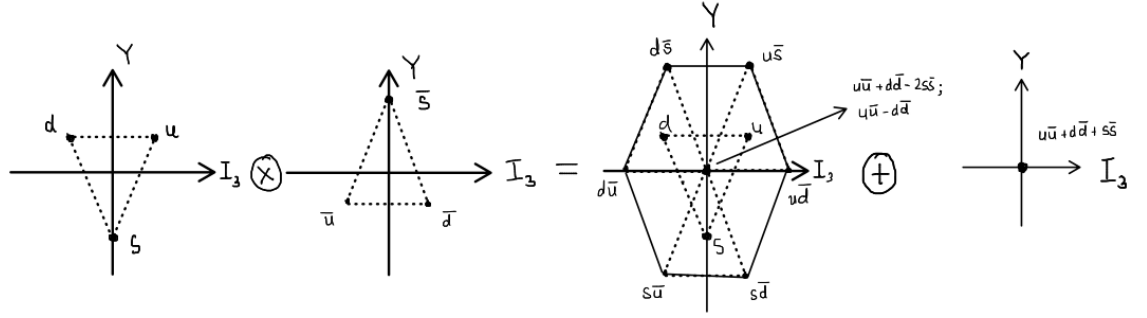




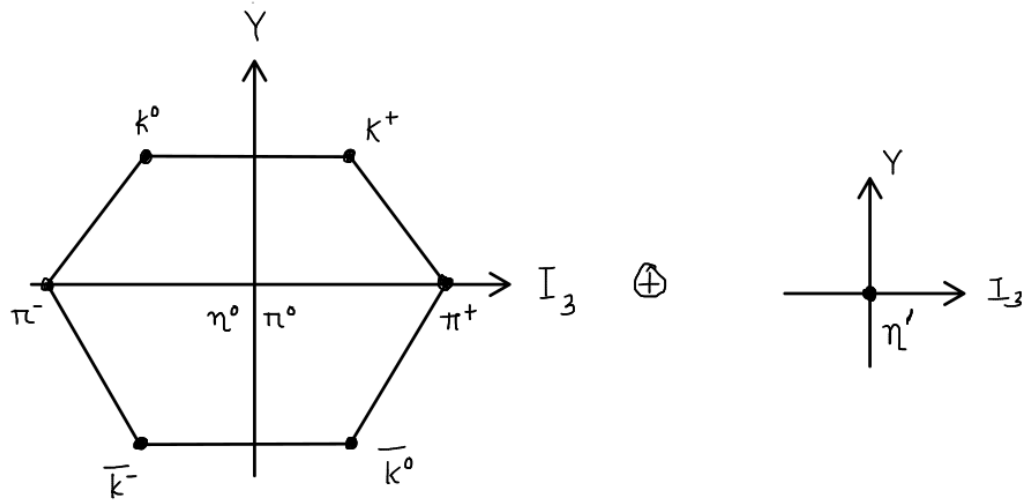
(b) $3 \otimes \bar{3}$.

Solution:

We know $3 \otimes \bar{3} = 8 \oplus 1$. Therefore weight diagrams will look like



With particles being



11. For two particles with isospin $I = 2$ and $I = \frac{1}{2}$, write down the states using Clebsch–Gordon decomposition.

(a) $|2, 2\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$ in terms of $\left| \frac{5}{2}, \frac{3}{2} \right\rangle$ and $\left| \frac{3}{2}, \frac{3}{2} \right\rangle$ states,

Solution:

$$\begin{aligned} \left| \frac{5}{2}, \frac{5}{2} \right\rangle &= |2, 2\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle \\ (\text{Operate by } J_- = J_{1-} + J_{2-}) \\ \left| \frac{5}{2}, \frac{3}{2} \right\rangle &= \frac{2}{\sqrt{5}} |2, 1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle + \frac{1}{\sqrt{5}} |2, 2\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \end{aligned} \quad (5)$$

The state $\left| \frac{3}{2}, \frac{3}{2} \right\rangle$ will be orthogonal to $\left| \frac{5}{2}, \frac{3}{2} \right\rangle$ and hence can be given by

$$\left| \frac{3}{2}, \frac{3}{2} \right\rangle = \frac{1}{\sqrt{5}} |2, 1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle - \frac{2}{\sqrt{5}} |2, 2\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \quad (6)$$

We can invert the two to obtain the desired state giving

$$\boxed{|2, 2\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \frac{1}{\sqrt{5}} \left| \frac{5}{2}, \frac{3}{2} \right\rangle - \frac{2}{\sqrt{5}} \left| \frac{3}{2}, \frac{3}{2} \right\rangle}$$

(b) $|2, -1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle$ in terms of $\left| \frac{5}{2}, -\frac{1}{2} \right\rangle$ and $\left| \frac{3}{2}, -\frac{1}{2} \right\rangle$ states,

Solution:

Applying J_- twice on (5) we get

$$\left| \frac{5}{2}, \frac{1}{2} \right\rangle = \sqrt{\frac{2}{5}} |2, -1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle + \sqrt{\frac{3}{5}} |2, 0\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \quad (7)$$

Applying J_- once on the state (6) we get

$$\left| \frac{3}{2}, -\frac{1}{2} \right\rangle = \sqrt{\frac{3}{5}} |2, -1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle - \sqrt{\frac{2}{5}} |2, 0\rangle \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \quad (8)$$

Now inverting these two we obtain

$$\boxed{|2, -1\rangle \left| \frac{1}{2}, \frac{1}{2} \right\rangle = \sqrt{\frac{2}{5}} \left| \frac{5}{2}, -\frac{1}{2} \right\rangle + \sqrt{\frac{3}{5}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle}$$

12. The $SU(3)$ generators satisfy

$$[F^a, F^b] = if^{abc} F^c.$$

We identify two diagonal generators F^3 and F^8 satisfying $[F^3, F^8] = 0$. Define

$$T^\pm = F^1 \pm iF^2, \quad U^\pm = F^6 \pm iF^7, \quad V^\pm = F^4 \pm iF^5,$$

with

$$T^3 = F^3, \quad Y = \frac{2}{\sqrt{3}} F^8.$$

They satisfy the commutation relations:

$$[T^3, T^\pm] = \pm T^\pm, \quad [Y, T^\pm] = 0,$$

$$[T^3, U^\pm] = \mp \frac{1}{2} U^\pm, \quad [Y, U^\pm] = \pm U^\pm,$$

$$[T^3, V^\pm] = \pm \frac{1}{2} V^\pm, \quad [Y, V^\pm] = \pm V^\pm,$$

$$[T^+, T^-] = 2T^3,$$

$$[U^+, U^-] = \frac{3}{2}Y - T^3,$$

$$[V^+, V^-] = \frac{3}{2}Y + T^3,$$

$$[T^+, V^-] = -U^-, \quad [T^-, U^+] = V^+, \quad [U^+, V^-] = T^-.$$

Determine all the non-vanishing structure constants f^{abc} using the above commutation relations.

Solution:

Along with the commutation we will use property that f^{abc} is completely antisymmetric for SU(3)

Proof:

by definition $f^{abc} = f^{acb}$

now

$$\begin{aligned} \text{Tr}([T^a, T^b]T^c) &= \text{Tr}(if^{abd}T^dT^c) \\ &= \frac{i}{2}f^{abd}\delta^{dc} \\ &= \frac{i}{2}f^{abc} \end{aligned}$$

$$\begin{aligned} \text{Tr}([T^a, T^b]T^c) &= -\text{Tr}([T^b, T^a]T^c) \\ &= -\frac{i}{2}f^{bac} \end{aligned}$$

Therefore $f^{abc} = -f^{bac}$

by using cyclic property of trace

$$\begin{aligned} \text{Tr}([T^a, T^b]T^c) &= \text{Tr}(T^a[T^b, T^c]) \\ &= -\text{Tr}(T^a[T^c, T^b]) \\ &= -\frac{i}{2}f^{cba} \end{aligned}$$

Therefore $f^{abc} = -f^{cba}$

Therefore *Structure constants for SU(3) are completely antisymmetric*

$$[F^3, F^1 + iF^2] = if^{31a}F^a - f^{32a}F^a = F^1 + iF^2$$

Comparing we get

$$f^{12\alpha} = \begin{cases} 1 & \text{for } \alpha = 3 \\ 0 & \text{otherwise} \end{cases}$$

$$[F^3, F^6 + F^7] = -\frac{1}{2}(F^6 + iF^7)$$

Therefore

$$f^{37\alpha} = \begin{cases} 1/2 & \text{for } \alpha = 6 \\ 0 & \text{otherwise} \end{cases}$$

$$[F^3, F^4 + iF^5] = if^{34a}F^a - f^{35a}F^a = \frac{1}{2}(F^4 + iF^5)$$

Therefore,

$$f^{34\alpha} = \begin{cases} 1/2 & \text{for } \alpha = 5 \\ 0 & \text{otherwise} \end{cases}$$

$$[2\sqrt{3}F^8, F^1 + iF^2] = 2\sqrt{3}(if^{81a}F^a - f^{82a}F^a) = 0$$

Therefore $f^{81a} = f^{82a} = 0 \quad \forall a$

$$\left[\frac{2}{\sqrt{3}}F^8, F^6 + iF^7 \right] = F^6 + iF^7$$

$$if^{86a}F^a - f^{87a}F^a = F^6 + iF^7$$

$$f^{86\alpha} = \begin{cases} \sqrt{3}/2 & \text{for } \alpha = 7 \\ 0 & \text{otherwise} \end{cases}$$

$$\left[\frac{2}{\sqrt{3}}F^8, F^4 + iF^5 \right] = F^4 + iF^5$$

$$if^{84a}F^a - f^{85a}F^a = F^4 + iF^5$$

$$f^{84\alpha} = \begin{cases} \sqrt{3}/2 & \text{for } \alpha = 5 \\ 0 & \text{otherwise} \end{cases}$$

Use $[T_+, V_-] = -U_-$

$$[F^1 + iF^2, F^4 - iF^5] = if^{14a}F^a + f^{15a}F^a - f^{24a}F^a + if^{25a}F^a = -F^6 + iF^7$$

$$f^{147} + f^{257} = 1 \quad (9)$$

$$f^{156} - f^{246} = -1 \quad (10)$$

This tells us that

$$f^{14\alpha} = 0 \quad \alpha \neq 7$$

$$f^{25\alpha} = 0 \quad \alpha \neq 7$$

$$f^{15\alpha} = 0 \quad \alpha \neq 6$$

$$f^{24\alpha} = 0 \quad \alpha \neq 6$$

Use $[T_-, U_+] = V_+$.

$$[F^1 - iF^2, F^6 + iF^7] = -F^4 - iF^5$$

which gives

$$f^{165} + f^{275} = -1 \quad (11)$$

$$f^{264} - f^{174} = -1 \quad (12)$$

$$[F^6 + iF^7, F^4 - iF^5] = F^1 - iF^2$$

This gives

$$f^{642} + f^{752} = -1 \quad (13)$$

$$f^{651} - f^{741} = 1 \quad (14)$$

solving (5),(6),(9) and (10) we get $f^{147} = f^{165} = f^{245} = f^{257} = 1/2$ Therefore the non-vanishing Structure constants are

$$\boxed{f_{123} = 1 \quad f_{458} = f_{678} = \sqrt{3}/2}$$

$$\boxed{f_{147} = f_{165} = f_{246} = f_{257} = f_{345} = f_{376} = 1/2}$$